

ARMEEN TAEB

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RESEARCH INTERESTS

My research objective is to broaden the scope of inferential methodology to non-standard settings. I leverage insights from statistical inference as well as concepts from discrete and continuous optimization, combinatorics, order theory, and causality.

ACADEMIC POSITIONS

- 2023- **Affiliate Faculty, eScience Institute**, University of Washington
- 2023- **Affiliate Faculty, Center for Statistics and the Social Sciences**, University of Washington
- 2022- **Assistant Professor, Department of Statistics**, University of Washington
- 2019-2022 **Foundations of Data Science Postdoctoral Associate**, ETH Zürich, Advisor: Peter Bühlmann

PROFESSIONAL POSITIONS

- 2017 **Data science intern**, Yahoo Inc.
- 2015 **Data science intern**, Jet Propulsion Laboratory, NASA
- 2012 **DSP Research intern: Baraniuk lab**, Rice University
- 2010-2011 **DSP Research intern: Hughes lab**, CU Boulder
- 2010 **Summer intern**, National Institute of Standards and Technology

EDUCATION

California Institute of Technology

PHD, ELECTRICAL ENGINEERING

Advisor: Venkat Chandrasekaran

Pasadena, CA

2015 - 2019

California Institute of Technology

M.S. IN ELECTRICAL ENGINEERING

Pasadena, CA

2013 - 2015

University of Colorado at Boulder

B.S. IN ELECTRICAL ENGINEERING AND APPLIED MATHEMATICS

Boulder, CO

2009 - 2013

PUBLICATIONS

UNDER REVIEW (†: indicates PhD advisee; * indicates equal contributions)

M. Valdez, A. Willis, **A. Taeb**, “Consensus tree estimation with false discovery control via partially ordered sets”

◇ 2026

A. Taeb*, R. Guo*, L. Henkel*, “Model-oriented graph distances via partially ordered sets”

◇ 2026

X. Zhang†, N. Keret, A. Shojaie, **A. Taeb**, “Convex mixed-integer programming for causal additive models with optimization and statistical guarantees”

◇ 2026, major revision at Journal of American Statistical Association

X. Zhang†, J. Bien, **A. Taeb**, “Quantifying uncertainty and stability among highly correlated predictors: a subspace perspective”

◇ 2026

J. Gamella, **A. Taeb**, C. Heinze-Deml, P. Bühlmann, “Characterization and greedy learning of Gaussian structural causal models under unknown interventions”

- ◇ 2026 – Minor revision at Journal of Machine Learning Research

JOURNAL PUBLICATIONS (* indicates equal contributions)

- R. Sergazinov, **A. Taeb**, I. Gaynanova, “[A spectral method for multi-view subspace learning using the product of projections](#)”
 ◇ 2025 – Biometrika
- T. Xu, S. Küçükyavuz, A. Shojaie, **A. Taeb**, “[An asymptotically optimal coordinate descent algorithm for learning Bayesian networks from Gaussian models](#)”
 ◇ 2025 – Journal of Machine Learning Research
- X. Shen, P. Bühlmann, **A. Taeb**, “[Causality-oriented robustness: exploiting general noise interventions](#)”
 ◇ 2025 – Journal of the American Statistical Association
- T. Xu*, **A. Taeb***, S. Küçükyavuz, A. Shojaie, “[Integer programming for learning directed acyclic graphs from non-identifiable linear models](#)”
 ◇ 2025 – Biometrika
- S. Engelke, **A. Taeb**, “[Extremal graphical modeling with latent variables via convex optimization](#)”
 ◇ 2025 – Journal of Machine Learning Research
- A. Taeb**, J. Gamella, C. Heinze-Deml, P. Bühlmann, “[Learning and scoring Gaussian latent causal models with unknown additive interventions](#)”
 ◇ 2024 – Journal of Machine Learning Research
- A. Taeb**, P. Bühlmann, V. Chandrasekaran “[Model selection over partially ordered sets](#)”
 ◇ 2024 – Proceedings of National Academy of Sciences
- Y. Chen, **A. Taeb**, and P. Bühlmann, “[A look at robustness and stability of \$\ell_1\$ vs. \$\ell_0\$ regularization: discussion of papers by Bertsimas et al. and Hastie et al.](#)”
 ◇ 2020 – Statistical Science
- A. Taeb**, P. Shah, and V. Chandrasekaran, “[False discovery and its control in low-rank estimation](#)”
 ◇ 2020 – Journal of the Royal Statistical Society, Series B
- A. Taeb**, P. Shah, and V. Chandrasekaran, “[Interpreting latent variables in factor models via convex optimization](#)”
 ◇ 2018 – Mathematical Programming
- A. Taeb**, J. Reager, M. Turmon, and V. Chandrasekaran, “[A statistical graphical model of the California reservoir network](#)”
 ◇ 2017 – Water Resources Research
- H. Qi, **A. Taeb**, and S. Hughes, “[Visual stylometry using background selection and wavelet- HMT-based Fisher Information distances for attribution and dating of impressionist paintings](#)”
 ◇ 2012 – EURASIP Signal Processing

BOOKS

- D. Sanz-Alonso, A. Stuart, and **A. Taeb**, “[Inverse problems and data assimilation](#)”
 ◇ 2023 – Cambridge University Press

PHD THESIS

- “[Latent-variable modeling: inference, algorithms, and applications](#)”
 ◇ 2019 – W. P. Carey & Co. Prize for outstanding thesis in Applied Mathematics

CONFERENCE PROCEEDINGS AND WORKSHOPS

- A. Taeb**, N. Ruggeri, C. Schnuck, F. Yang, “[Provable concept learning for interpretable predictions via variational inference](#)”
 ◇ 2022 – ICML workshop on AI4Science
- A. Taeb**, A. Maleki, C. Studer, and R. Baraniuk, “[Maximin analysis of message passing algorithms for block sparse signals](#)”
 ◇ 2013 – Signal Processing with Adaptive Sparse Structured Representations (SPARS)

TECHNICAL REPORTS

- M. Azadkia, **A. Taeb**, P. Bühlmann, “[A fast non-parametric approach for local causal structure learning](#)”
 ◇ 2022 – arXiv 2111.14969
- A. Taeb**, P. Shah, and V. Chandrasekaran, “[Learning exponential family graphical models with latent variables using regularized conditional likelihood](#)”
 ◇ 2020 – arXiv 2010.09386

GRANTS

“2025 conferences for new researchers in statistics, probability, and data science”

- ◇ CO-PI: with E. Lila
- ◇ National Science Foundation, DMS, Statistics; 40k

“False discovery control in non-standard settings”

- ◇ National Science Foundation, DMS, Statistics; 225k

“False discovery control for causal structure learning”

- ◇ 2023-2024; Royalty Research Fund, University of Washington; \$40k

“Interpretable predictions for medical imaging diagnostics”

- ◇ CO-PI: with F. Yang, J. Vogt, and E. Ozkan
- ◇ 2021-2023; Hasler Foundations, Switzerland; CHF 510k

AWARDS & HONORS

2019-2021	Foundations of Data Science Postdoctoral Fellowship, ETH
2020	W. P. Carey & Co. Prize for outstanding thesis in Applied Mathematics, Caltech
2016-2018	Resnick Institute Fellowship for Sustainability Research, Caltech
2013-2014	Electrical Engineering Graduate Fellowship, Caltech
2013	GRFP Honorable Mention, NSF
2013	Distinguished Senior in Electrical Engineering, University of Colorado at Boulder

TEACHING & MENTORING

TEACHING

Winter 2025	STAT 416, Introduction to Machine Learning, Undergraduate Course, 137 Students	UW
Wi. 2026	STAT 538, Statistical Learning, Graduate Course, 28 Students	UW
Spr 2025	STAT 435, Introduction to Statistical Machine Learning, Undergraduate Course, 83 Students	UW
Wi. 2025	STAT 538, Statistical Learning, Graduate Course, 28 Students	UW
Wi. 2025	STAT 528, Applied Statistics Capstone, Graduate Course, 32 Students	UW
Wi. 2024	STAT 538, Statistical Learning, Graduate Course, 18 Students	UW
Wi. 2024	STAT 528, Applied Statistics Capstone, Graduate Course, 38 Students	UW
Fa. 2023	STAT 591, Multiple Testing and Modern Inference, Graduate Course, 12 Students	UW
Sp. 2023	BIOST 558, Statistical Machine Learning for Data Scientists, Graduate Course, 57 Students	UW
Wi. 2023	STAT 538, Statistical Learning, Graduate Course, 4 Students	UW
Wi. 2023	STAT 528, Applied Statistics Capstone, Graduate Course, 36 Students	UW
Fa. 2020	Seminar on Multiple Testing for Modern Data Science, Co-Instructor (with Matthias Löffler)	ETH
Fa 2018	Inverse Problems & Data Assimilation, Co-instructor (with Andrew Stuart)	UDSI

MENTORING

PhD students and postdocs

2023-	Xiaozhu Zhang, PhD advisee	UW
2023-	Tong Xu, Advisor: Simge Küçükyavuz	Northwestern
2025-	Maria Valdez, Advisor: Amy Willis	UW

Masters students

2023-2025	Bojun Chen, advisee, first position: PhD at UW Biostatistics	UW
2022	Felix Hafenmair, thesis advisee	ETH
2022	Zipei Geng, thesis advisee, first position: PhD at Kaust	ETH
2021	Carina Schnuck, thesis advisee, first position: Citi bank	ETH
2020-2021	Juan Gamella, thesis advisee, first position: PhD at ETH	ETH

Undergraduate students

2025-	Jian Kang, research advisee	UW
2025-	Arnav Mazumder, research advisee	UW
2025-	Carter Lembo, research advisee	UW

ACADEMIC SERVICE & LEADERSHIP

SCIENTIFIC EVENTS AND ORGANIZATION

IMS New Researchers

- 2024-2026 President of IMS New Researchers Group, IMS
- 2026 Co-organizer of the first Asian New Researchers Conference, Hong Kong
- 2025 Co-organizer of the first European New Researchers Conference, Spain
- 2024-2026 Co-organizer of the 24-26th IMS New Researchers Conference, USA

Conference Session organizer

- 2026 IMS Annual Meeting, topic: “Modern optimization for statistical learning”.
- 2025 Joint Statistical Meeting, topic: “The interplay of statistics, optimization, and geometry for modern data science”.
- 2024 Co-organizer of a contributed session in IMS-Bernoulli Conference in Probability and Statistics, Co-organized with Mateo Diaz, topic: “False discovery control in non-standard settings”.
- 2024 Co-organizer of invited session in American Causal Inference Conference, Co-organized with Ali Shojaie, topic: “Optimization methods for causal discovery”.
- 2023- Organizer: IMS New Researcher Group Seminar Series, Organize talks from young researchers in data science.
- 2023 Co-organizer of minisymposium in SIAM Conference on Optimization, Co-organized with Xinwei Shen, topic: “Application of optimization for causal discovery”.
- 2020-2022 Co-organizer: Young Data Science Seminar Zürich, Organize talks from young researchers.
- 2017 Co-organizer: Co-organizer of minisymposium in SIAM Conference on Optimization, Co-organized with Yong Sheng Soh, topic: “Lift-and-project methods for data science”.

REVIEWING

Journals: Annals of Statistics, Bernoulli, Biometrika, Electronic Journal of Statistics, Journal of American Statistical Association, Journal of Machine Learning Research, Journal of Royal Statistical Society Series B, SIAM Journal of Mathematics of Data Science, Statistical Science.

Conferences: Neurips, UAI.

SELECTED TALKS

“COMPLEX MODEL SELECTION VIA POSETS: CORRELATED FEATURES, CAUSAL GRAPHS, AND PHYLOGENETIC TREES”

- ◊ Statistical Foundations of Artificial Intelligence, November 2026
- ◊ Statistical Learning and Data Science, November 2026
- ◊ IMS Annual meeting, July 2026,
- ◊ IMS Asia Pacific Rim (Hong Kong), June 2026
- ◊ American Causal Inference Conference, May 2026
- ◊ Department of Statistics, University of Chicago, October 2025

“INTEGER PROGRAMMING FOR LEARNING DIRECTED ACYCLIC GRAPHS”

- ◊ International Conference on Frontiers of Data Science (Hangzhou, China), July 2024
- ◊ Pacific Causal Inference Conference (Shanghai, China), July 2024
- ◊ American Causal Inference Conference, May 2024

“MODEL SELECTION OVER PARTIALLY ORDERED SETS”

- ◊ Department of Statistics, University of Melbourne, December 2024
- ◊ Department of Statistics, Rutgers University, November 2024
- ◊ Bernoulli-IMS World Congress in Probability and Statistics, August 2024
- ◊ Marshall Statistics Seminar, University of Southern California, April 2024
- ◊ eScience Institute, University of Washington, April 2024
- ◊ International Conference on Selective Inference, March 2024
- ◊ Department of Statistical Science, Duke University, December 2023
- ◊ Department of Biostatistics, University of Washington, November 2023

“QUANTIFYING RARE EVENTS WITH EXTREMAL GRAPHICAL MODELS”

- ◊ Department of Atmospheric Sciences, University of Washington, May 2024

“CAUSALITY AND ROBUSTNESS FROM HETEROGENEOUS DATA”

- ◊ Interactive Causal Learning Conference, Boca Raton, December 2023
- ◊ Department of Industrial and Systems Engineering, University of Washington, May 2023

- ◊ Department of Economics, University of Washington, May 2023

“PROVABLE CONCEPT LEARNING FOR INTERPRETABLE PREDICTIONS”

- ◊ Department of Medicine, University of California at San Francisco, November 2022

“CAUSAL STRUCTURE LEARNING WITH UNKNOWN NOISE INTERVENTIONS”

- ◊ SIAM Conference on Optimization, June 2023
- ◊ American Causal Inference Conference, May 2023
- ◊ Department of Statistics, University of Washington, October 2022
- ◊ COMSTAT, August 2022

“PERTURBATIONS AND CAUSALITY IN GAUSSIAN LATENT VARIABLE MODELS”

- ◊ CMStatistics, December 2021
- ◊ Online Causal Inference Seminar, June 2021
- ◊ SIAM Conference on Algebraic Geometry, August 2021
- ◊ SIAM Conference on Optimization, July 2021
- ◊ Seminar for Statistics and Data Science, TU München, March 2021

“FALSE DISCOVERY AND ITS CONTROL IN LOW-RANK ESTIMATION”

- ◊ SIAM Conference on Optimization, May 2021
- ◊ Statistics@UPF Seminars, March 2020
- ◊ SIAM Conference of Algebraic Geometry, June 2019
- ◊ Department of Computer Science, Northeastern, December 2018
- ◊ Laboratory for Information and Decision Systems, MIT, November 2018
- ◊ Workshop on New Signal Models and their Information Content, Banff, November 2018
- ◊ Statistics Seminar, University of Chicago, October 2018
- ◊ Seminar For Statistics: ETH Zürich, September 2018

“FROM DATA SCIENCE TO HYDROLOGY, CALIFORNIA RESERVOIRS DURING DROUGHT”

- ◊ Wonderful Company HQ, July 2018
- ◊ RAND Corporation, June 2018
- ◊ International Congress on Environmental Modelling and Software, May 2018
- ◊ San Francisco Water Public Utilities Commission, March 2018

“INTERPRETING LATENT VARIABLES VIA CONVEX OPTIMIZATION”

- ◊ Allerton Conference on Communication, Control, and Computing, October 2017
- ◊ SIAM Optimization, May 2017